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aPhyloGeo: a multi-platform Python package for analyzing phylogenetic trees with climatic parameters

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Abstract

aPhyloGeo is a Python library that explores the relationship between species evolution and environmental pressures, particularly climate. By integrating genetic and climatic data, it helps researchers investigate evolutionary adaptations and identify genetic regions influenced by environmental factors.

The core feature of the software is a comprehensive phylogenetic analysis pipeline with three investigation levels: 1) genetic relationships, 2) climatic impact assessment, and 3) biogeographic correlations. This approach facilitates understanding how species adapt to their environments.

aPhyloGeo employs robust algorithms to ensure statistically sound correlations, providing flexibility through its modular structure and promoting collaboration through its open-source nature.

Available as a PyPI package, *aPhyloGeo* enhances understanding of evolutionary processes and informs conservation efforts, helping prioritize species and populations for preservation in the face of climate change.

Statement of Need

Phylogeography, the study of the interplay between evolution and environment, has long been hindered by the challenges of integrating complex phylogenetic and environmental datasets. aPhyloGeo contributes to this field by offering a user-friendly Python package that streamlines phylogeographic analysis. By facilitating the investigation of the intricate relationship between genetic variation and environmental factors, aPhyloGeo is poised to advance our understanding of evolution, biogeography, and conservation efforts. This powerful package makes phylogeographic analysis more accessible and efficient, opening new avenues for research and discovery. The aPhyloGeo package is designed for versatile integration, offering flexibility to be incorporated into various environments. It can be seamlessly embedded within Python scripts for custom workflows, deployed on cloud platforms for scalable analysis, or integrated into web and desktop applications for user-friendly access. This adaptability makes it a valuable tool for diverse research and development needs, from individual projects to large-scale applications.

Related Software Packages

While BEAST (Drummond & Rambaut, 2007) and MrBayes (Ronquist et al., 2012) are popular for phylogenetic analyses, they lack integrated tools for analyzing climatic data. Other software like ArcGIS (Scott & Janikas, 2009) focuses on geospatial analysis but may not be tailored for phylogeographic research. aPhyloGeo bridges this gap by providing a unified platform for both types of analysis, with a streamlined workflow.

aPhyloGeo: A Software Solution

aPhyloGeo is a software package designed for detailed phylogeographic analysis, considering both genetic and climatic data.

Addressing Critical Questions

Given the urgency of climate change, new bioinformatics tools are needed. *aPhyloGeo* will help researchers address critical questions like:

- How has past climate shaped current genetic diversity and distribution?
- How will species distributions evolve under future climates?
- Which populations are most vulnerable to extinction?
- Are there genetic signatures of local adaptation?

Contribution and Impact

aPhyloGeo aims to provide robust tools for analyzing and interpreting phylogeographic data. *aPhyloGeo* will help unravel the complex interplay between genetics, geography, and environment, informing conservation strategies and predicting impacts of climate change.

Pipeline

Figure 1 outlines the key steps for conducting phylogeographic analysis using *aPhyloGeo*. For a more detailed understanding, please refer to the comprehensive tutorial and provided with the software.

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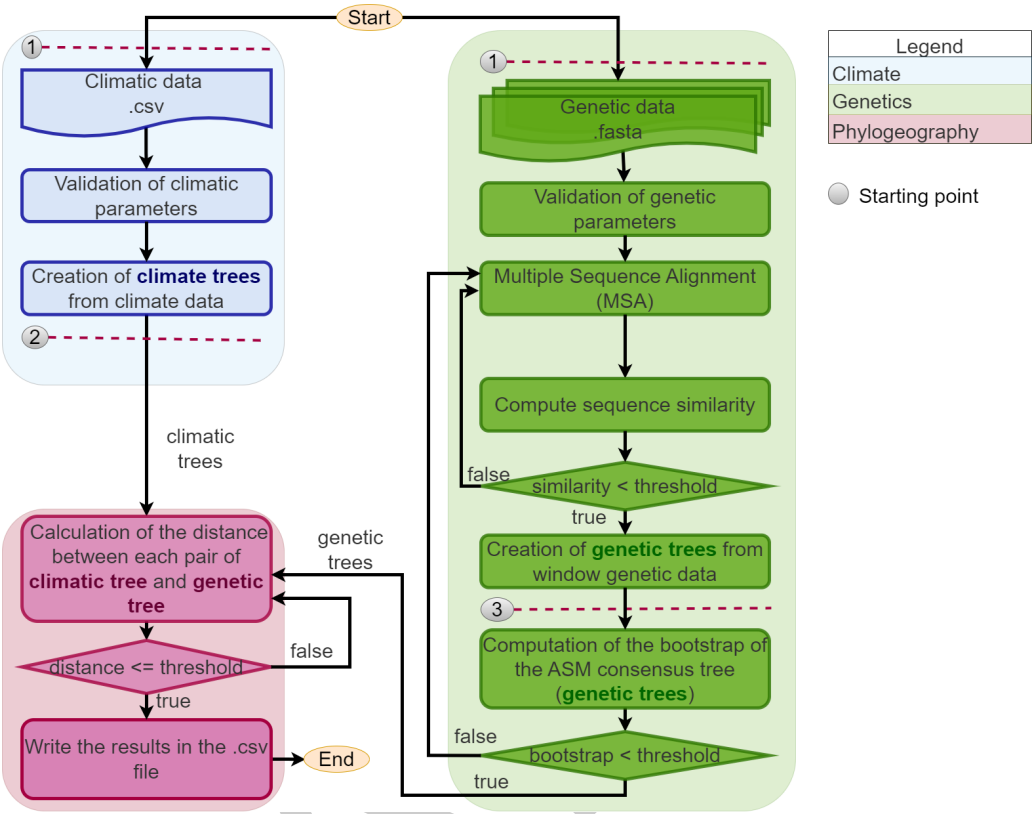

Figure 1: The workflow of the algorithm. The operations within this workflow include several blocks.

Multiprocessing

The algorithm supports multiprocessing, allowing the simultaneous analysis of multiple sliding windows (i.e., corresponds to the alignment sub-sequence where the starting point is located after the first position of the alignment, and the ending point is located before the last position of the alignment) within genetic data. This feature is particularly recommended for large datasets, as it significantly speeds up the analysis by dividing the input sequences into smaller chunks that can be processed in parallel.

Dependencies

This work relies on the following main software packages (refer to pyproject.toml file):

- ete3 (Huerta-Cepas et al., 2016).
- Bio (Cock et al., 2009).

Methods

The methods validate input parameters to ensure data accuracy and prevent errors. These parameters, such as file names, specimen identifiers, bootstrap thresholds, distance methods, and alignment methods, are detailed in the YAML configuration file (refer to the YAML file).

Tree Comparison

In the comparison of phylogenetic trees, which are constructed based on genetic data, with climatic trees, a crucial step involves applying a phylogeography approach. Climatic trees are

novel constructs that represent the evolutionary relationships among species or populations based on climatic parameters instead of genetic data. This methodology, developed by our group, allows for the assessment of how environmental factors, particularly climate, influence the evolutionary trajectories of species.

Climatic trees are constructed by analyzing climatic data associated with the geographical locations of the species or populations under study. By using algorithms that correlate climatic variables with genetic data, these trees help identify patterns of adaptation to environmental pressures. This method provides a unique perspective on the evolutionary processes driven by climate, offering insights that are not possible through genetic data alone.

Tree Comparison Methodology

In comparing these trees, aPhyloGeo uses the Robinson and Foulds distance for topology evaluation and the Least Squares distance for assessing branch length differences. This dual approach ensures that both the structural and quantitative aspects of the evolutionary relationships are rigorously analyzed, providing a comprehensive understanding of the interplay between genetics and climate.

Spatial Proximity Confounding Effects

An important limitation of the current *aPhyloGeo* analysis pipeline is the potential confounding effects of spatial proximity on both genetic relatedness and climate similarity. Spatial autocorrelation can introduce biases, as geographically close samples may exhibit similar genetic and climatic characteristics due to their proximity. Future versions of *aPhyloGeo* could address this by incorporating spatial autocorrelation analysis and spatial regression models.

Editing Multiple Sequence Alignment Methods

Multiple Sequence Alignment (MSA) holds immense significance in bioinformatics as it serves as a foundational step for the comparison and analysis of biological sequences. Here is an in-depth overview of some widely used MSA methods: 1) **Pairwise Alignment** (H. Li, 2018), 2) **MUSCLE** (Edgar, 2004), 3) **CLUSTALW** (Hung & Weng, 2016), and 4) **MAFFT** (Katoh & Standley, 2013).

Similarity Methods

Sequences with notable variability are specifically retained for analysis. The dissimilarity assessment between each sequence pair involved the application of an extensive set of 8 metrics: 1) **Hamming distance** (Labib et al., 2019), 2) **Levenshtein distance** (Yujian & Bo, 2007), 3) **Damerau-Levenshtein distance** (Zhao & Sahni, 2019), 4) **Jaro similarity** (Pradhan et al., 2015), 5) **Jaro-Winkler similarity** (Pradhan et al., 2015), 6) **Smith–Waterman similarity** (Waterman & Smith, 1978), 7) **Jaccard similarity** (Bag et al., 2019), and 8) **Sørensen-Dice similarity** (X. Li et al., 2020).

Conclusion

The *aPhyloGeo* pipeline provides an integrative framework for advanced co-analysis of genetic and climatic data, promoting simplicity and reproducibility. Future enhancements include clustering techniques and improved alignment methods, alongside novel metrics like quartet and bipartition distances for better genetic diversity assessment. By emphasizing software development best practices and open-source principles, *aPhyloGeo* ensures reliability and sets the stage for ongoing innovation. Notably, projects such as iPhyloGeo++ demonstrate the practical application and impact of aPhyloGeo in phylogeographic research. This underscores its utility and potential for broad adoption in the research community. By prioritizing user

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empowerment and comprehensive analysis, it aims to advance phylogeographic research and deepen our understanding of species-environment interactions.

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